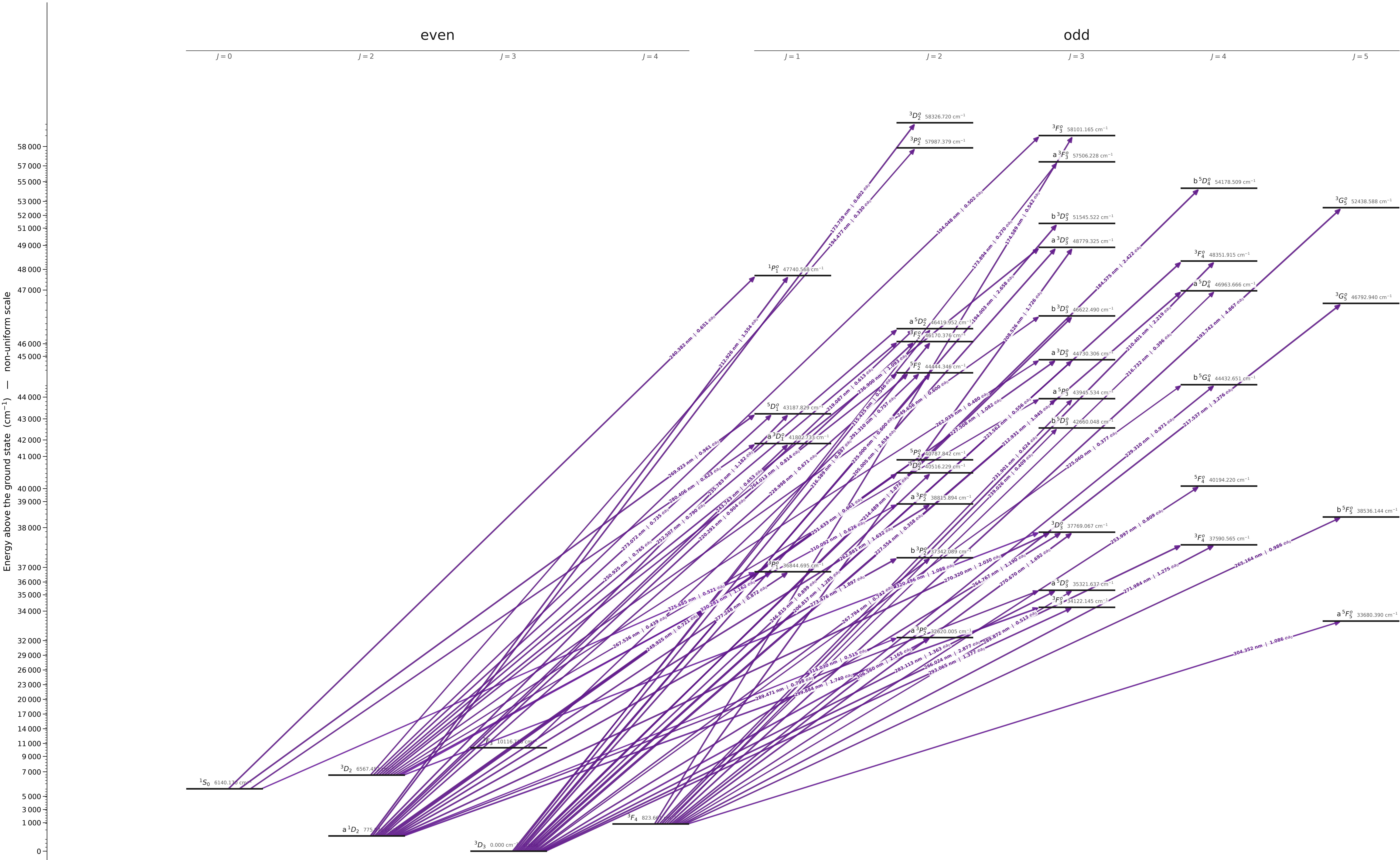
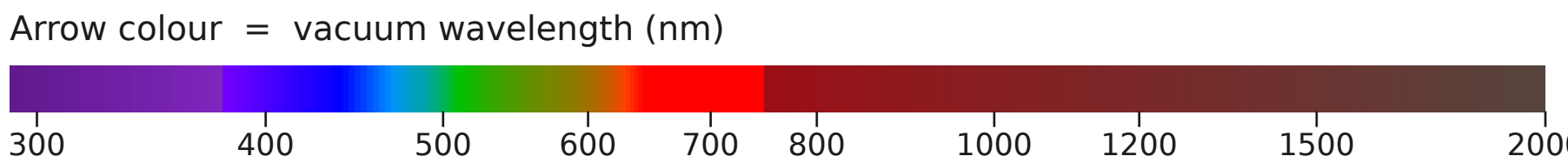


Platinum

Pt (Pt I). Strongest classified transitions below 2 μm (NIST): 41 levels, 70 lines.

Pt⁺ ionisation limit 72257.80 cm⁻¹ = 8.95883 eV (138.393 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||e r||U'|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨U e r U'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
³ D ₃ – b ⁵ D ₂ ^o	184.5750	-	1624.2308	2.422	2.100e+08	-	-	-
³ F ₄ – ³ G ₅ ^o	193.7424	-	1547.3766	4.867	6.000e+08	-	-	-
³ D ₃ – b ³ D ₃ ^o	194.0033	-	1545.2959	2.658	2.800e+08	-	-	-
³ D ₃ – a ³ D ₂ ^o	205.0049	204.9392	1462.3674	2.634	2.330e+08	-	-	-
³ F ₄ – a ³ D ₃ ^o	208.5259	208.4596	1437.6746	1.726	9.510e+07	-	-	-
³ F ₄ – ³ F ₄ ^o	210.4012	210.3345	1424.8612	2.219	1.190e+08	-	-	-
a ¹ D ₂ – ¹ P ₁ ^o	212.9259	212.8587	1407.9660	1.554	1.690e+08	-	-	-
³ D ₃ – a ⁵ D ₂ ^o	212.9306	212.8634	1407.9353	1.945	8.820e+07	-	-	-
³ D ₃ – b ³ D ₂ ^o	214.4888	214.4212	1397.7071	1.874	1.030e+08	-	-	-
³ F ₄ – ³ G ₅ ^o	217.5366	217.4684	1378.1243	3.276	1.920e+08	-	-	-
a ¹ D ₂ – ⁵ D ₂ ^o	235.7826	235.7105	1271.4784	1.182	7.200e+07	-	-	-
a ¹ D ₂ – a ³ F ₂ ^o	262.8811	262.8027	1140.4110	1.632	5.940e+07	-	-	-
³ D ₃ – ³ F ₄ ^o	266.0242	265.9451	1126.9368	2.877	9.900e+07	-	-	-
a ¹ D ₂ – ³ D ₃ ^o	270.3200	270.2399	1109.0280	2.030	6.040e+07	-	-	-
a ¹ D ₂ – b ³ P ₂ ^o	273.4765	273.3956	1096.2275	1.897	7.130e+07	-	-	-
³ D ₃ – a ³ P ₂ ^o	306.5603	306.4713	977.9231	2.165	6.590e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).